

Thesis Mid Report

"Breast Cancer Detection From Whole Slide Images Using Deep Learning"

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Abstract

Breast cancer remains one of the most prevalent and life-threatening diseases worldwide, with millions of women affected annually. While whole-slide imaging (WSI) has become an essential component in histopathological analysis, the manual interpretation of these images is labor-intensive and prone to inter-observer variability. Current deep learning approaches, despite showing promise, struggle with the vast size of WSIs, spatial inconsistencies and rely on sparsely annotated regions, resulting in the loss of global tissue context. This study proposes a parallel feature learning framework that integrates Convolutional Neural Networks (CNNs) for fine-grained cellular detail extraction and Vision Transformers (ViTs) for capturing long-range dependencies. This approach enables the comprehensive classification of biomarkers ER, PR, and HER2 within a unified pipeline, improving diagnostic accuracy and biomarker interaction modeling. By optimizing feature fusion and enhancing shared learning, the proposed framework improves generalization and provides a more effective AI-driven solution for breast cancer diagnosis. By maintaining both microscopic and macroscopic tissue features in automated analysis, this research aims to enhance the accuracy of breast cancer detection and treatment planning, ultimately contributing to improved patient outcomes.

Chapter 1

1.1 Introduction

Breast cancer is one of the most common cancers among women. According to the World Health Organization (WHO), 2.3 million women were diagnosed with breast cancer and 670000 deaths all over the world in 2022 [1]. The disease originates in the cells of the breast tissue, with uncontrolled cell growth leading to the formation of malignant tumors. Various factors contribute to the development of breast cancer, including genetic predisposition, hormonal imbalances, age, and lifestyle choices. Early detection significantly improves patient prognosis, emphasizing the importance of advanced diagnostic techniques. Histopathological analysis remains a cornerstone of breast cancer diagnosis, playing an essential role in identifying tumor characteristics and guiding treatment decisions.

1.1.1 Background/Domain Concepts

Histopathological examination involves the microscopic evaluation of tissue samples obtained through biopsy procedures. These samples are typically stained with specialized dyes, such as Hematoxylin and Eosin (H&E) and Immunohistochemistry (IHC) markers, to highlight cellular structures and specific biomarkers. Pathologists carefully analyze these stained slides to identify cancerous cells, determine their characteristics, and assess the extent of tumor invasion. Critical biomarkers in breast cancer include Estrogen Receptor (ER), Progesterone Receptor (PR), and Human Epidermal Growth Factor Receptor 2 (HER2). The Allred scoring system is commonly employed to assess ER and PR status, combining the proportion of positive cells and staining intensity to yield a score ranging from 0 to 8. HER2 status is evaluated based on the intensity and completeness of membrane staining, with scores of 0 and 1 considered negative, 2 equivocal, and 3 positive [2]. Equivocal cases often require further testing, such as fluorescence in

situ hybridization (FISH), to determine HER2 gene amplification [3]. However, the traditional manual approach to histopathological evaluation presents several challenges. The process is time-consuming, labor-intensive, and susceptible to subjective variability among pathologists.

1.1.2 The Problem

Current deep learning approaches for breast cancer detection in whole-slide imaging (WSI) are constrained by inherent architectural and analytical gaps. Convolutional Neural Networks (CNNs), while proficient at extracting localized cellular features such as nuclear morphology and staining intensity, lack the capacity to model long-range dependencies in tissue architecture due to their localized receptive fields [2]. Conversely, Vision Transformers (ViTs) excel at capturing global contextual patterns through self-attention mechanisms but often overlook fine-grained cellular details critical for precise biomarker quantification [3]. When deployed sequentially where CNNs process local features first, followed by ViTs analyzing global structures the resulting disjointed workflow disrupts spatial coherence, leading to misaligned tissue representations and diminished diagnostic reliability [4].

Furthermore, existing methodologies predominantly focus on isolated biomarker analysis, such as HER2 or ER, neglecting clinically significant interactions between ER, PR, and HER2. These challenges highlight the necessity for an integrated framework that harmonizes localized and global feature extraction while enabling multi-biomarker interaction modeling, bridging cellular precision with tissue-wide context to meet clinical diagnostic demands.

1.1.3 Motivation

The motivation for this research stems from the need to develop an integrated AI-driven framework that can automate the entire histopathological analysis process. By dividing whole-slide images into overlapping patches and employing a parallel feature learning approach, this study aims to enhance biomarker classification by simultaneously capturing fine-grained cellular details and global tissue structures. Integrating CNNs for local feature extraction and ViTs for long-range dependencies will preserve spatial relationships and improve biomarker interaction modeling. This unified multi-biomarker classification system is expected to enhance diagnostic accuracy, streamline clinical workflows, and support more precise treatment decisions in breast cancer care.

1.1.4 Scope

This research aims to develop an AI-driven framework for histopathological analysis of breast cancer, focusing on integrating biomarker evaluation and Allred scoring into a unified pipeline. By Focusing on advancements in AI-driven histopathological analysis for breast cancer diagnostics from 2023 to 2025, and excluding genomic-based detection methods, radiological imaging, and non-histopathological diagnostics, this study seeks to address current limitations and pave the way for more effective and scalable solutions that benefit pathologists, healthcare professionals, and patients alike.

1.2 Research Objectives

This research aims to address the limitations in AI-driven histopathological analysis for breast cancer diagnostics. The key objectives are:

1. Integrate Local and Global Feature Extraction: Combine Convolutional Neural Networks (CNNs) and Vision Transformers (ViTs) to simultaneously capture fine-grained cellular details and overarching tissue structures from Whole Slide Images (WSIs).
2. Develop a Unified Multi-Biomarker Model: Create a comprehensive framework that evaluates multiple biomarkers concurrently, enhancing diagnostic accuracy through shared feature learning.
3. Preserve Spatial Relationships: Ensure the model maintains spatial relationships within WSIs by processing local and global features in parallel, rather than sequentially.
4. Ensure that the research outcomes directly contribute to improving breast cancer diagnostics by providing a comprehensive and automated solution for biomarker assessment. This solution will aid pathologists in making more accurate and consistent clinical decisions.
5. Complete the development, training, and evaluation of the proposed solution by 2025. Intermediate milestones will include model development, testing, and optimization phases to ensure timely progress and completion.

Chapter 2

2. Literature Review

2.1. Biomarker Classification in Breast Cancer

Advancements in deep learning have significantly contributed to the automation of histopathological analysis for breast cancer detection. Several studies have explored the use of convolutional neural networks (CNNs), Vision Transformers (ViTs), and hybrid models for biomarker assessment and classification in whole-slide imaging (WSI).

Hasan et al. [2] developed a real-time segmentation and classification model to assist pathologists in breast cancer diagnostics. Their study incorporated SegNet and U-Net architectures with modified decoders that utilized ResNet, VGG, and DenseNet backbones. The model demonstrated high segmentation performance with an F1-score of 0.902 on a private dataset and 0.903 on the MoNuSeg dataset, while classification achieved an F1-score of 0.833. However, HER2 biomarker analysis was not included in the training or testing phase, limiting the model's applicability for comprehensive breast cancer diagnosis. Additionally, the model struggled with long-range dependencies, suggesting that advanced mechanisms such as attention-based models could further enhance performance.

Rehman et al. [3]proposed a deep-learning framework to classify Normal, Amplified, and Non-Amplified regions in HER2-SISH slides. Their methodology employed a two-stage classification approach, starting with patch-level classification followed by whole-slide regional classification. Among the models tested, Vision Transformer (ViT) outperformed

others, achieving a patch-level accuracy of 99.9%. However, the generalization accuracy on whole-slide images was significantly lower (78.8%), indicating difficulties in integrating local and global tissue information. The study also did not consider other critical biomarkers such as ER and PR, which are necessary for a more comprehensive breast cancer characterization.

Saranyaraj et al.[5] focused on early breast cancer detection by classifying HER2 and ER biomarkers using histopathological images. The study employed a Stacked Sparse Autoencoder (SSAE) for feature extraction and a deep Convolutional Neural Network (DCNN) for classification, achieving a nuclei classification accuracy of 99.73% and a HER2/ER patch level classification accuracy of 97.52%. However, the model did not incorporate the PR biomarker, which is critical for hormone receptor status assessment. Additionally, the lack of external validation raises concerns about the model's generalizability, and the high computational cost of the SSAE approach could limit its practical deployment in real-time clinical settings.

Che et al. (2023)[6] developed an automated method for HER2 immunohistochemistry (IHC) scoring in whole-slide images using deep learning. Their approach involved threshold-based segmentation for region of interest (ROI) extraction, followed by tumor segmentation and patch-wise classification using ResNet34. The model achieved a high patch-level precision of 95.77%. However, the study focused solely on HER2, omitting ER and PR biomarkers, which are essential for a comprehensive pathological assessment. Furthermore, the reliance on threshold-based segmentation could introduce variability in the detection process, potentially impacting model robustness.

Shamai et al. [7] investigated the use of deep learning to predict ER, PR, and HER2 receptor status from H&E-stained slides. The study trained separate models for each biomarker using convolutional neural networks (CNNs) for local feature extraction, followed by a global feature aggregation step to capture overall tissue architecture. The model achieved high area-under-the-curve (AUC) scores of 0.955 for ER, 0.807 for PR, and 0.846 for HER2. However, the study did not used shared feature learning across receptor types, limiting its ability to capture multi-scale relationships between biomarkers. The late fusion strategy used for feature aggregation restricted the model's adaptability, emphasizing the need for parallel processing techniques that better preserve spatial relationships in histopathological images.

Goyal et al. [4] developed a deep learning model to predict breast cancer recurrence risk using whole-slide images and clinicopathological data as an alternative to the Oncotype DX test. Their approach combined CNN-based local feature extraction with Vision Transformer-based global feature extraction. While this hybrid model provided valuable insights into broad tumor patterns, it processed local and global features sequentially, potentially leading to the loss of spatial relationships between cell-level and tissue-wide structures. Additionally, the study focused on recurrence risk prediction rather than direct biomarker classification, which limits its direct applicability to breast cancer subtyping.

Mujawar et al.[8] aimed to enhance breast cancer classification in whole-slide images using stacked convolutional neural networks (CNNs). Their study employed a patch-based classification approach, initially using small patches (224×224 pixels) and later integrating larger patches (512×512, 768×768, 1024×1024) for improved accuracy. The model achieved 90% accuracy in binary classification but only 76.6% accuracy in three-

class classification, highlighting challenges in capturing contextual information from small patches. Additionally, the computational cost increased significantly for larger patch sizes, limiting the scalability of the approach for high-resolution whole-slide images.

Aboudessouki et al. [9] proposed a deep learning pipeline for automated HER2 classification in breast cancer histology images. Their methodology involved extracting 128×128 pixel patches from WSIs, applying Otsu thresholding for preprocessing, and using DenseNet for HER2 classification at both patch-wise and pixel-wise levels. The model achieved a high accuracy of 0.97 for patch-wise classification but lower accuracy (0.79) for pixel-wise classification, indicating potential limitations in fine-grained biomarker assessment. Additionally, the model was evaluated on a small dataset of only 10 cases, raising concerns about its generalizability. Moreover, the pipeline focused exclusively on HER2, lacking a comprehensive molecular subtyping model that considers ER and PR biomarkers.

While these studies demonstrate the potential of deep learning for breast cancer detection and biomarker classification, they also reveal key limitations. CNN-based approaches struggle with capturing long-range dependencies due to their localized receptive fields, limiting their ability to integrate global tissue structure in whole-slide images. In contrast, ViT-based models excel at capturing global dependencies but often lose fine-grained cellular details due to patch-wise processing, making them computationally expensive. Hybrid models attempt to bridge this gap but typically integrate local and global features sequentially rather than interactively, leading to suboptimal feature representation. Moreover, most studies focus on individual biomarkers rather than considering all key biomarkers in a unified pipeline, limiting their clinical applicability for comprehensive breast cancer assessment.

Author(s) & Year	**Objective**	**Methodology**	**key Findings**	**Limitations**	**Relevance**
Hasan et al (2024)[2]	Develop a real-time segmentation and classification model for breast cancer histopathology images to assist pathologists	**(1)** Modified SegNet and U-Net decoders with ResNet, VGG, DenseNet backbones. **(2)** Real-time processing with GUI interface	Segmentation F1-score: 0.902 (private), 0.903 (MoNuSeg dataset), Classification F1-score: 0.833 (private)	**(1)**HER2 Not used in training or testing. **(2)**Limited ability to model long-range dependencies. **(3)**Lower classification accuracy highlights the need for advanced mechanisms like attention models for improvement	Provides insights into real-time segmentation, classification pipelines and architectural enhancements with decoders, though lacking HER2 analysis, a critical focus in our research.
Rehman et al. (2024) [3]	Develop a deep-learning approach to classify Normal, Amplified, and Non-Amplified regions in HER2-SISH slides	Two-stage approach: **(1)** Patch-level classification on annotated regions **(2)** WSI-level regional classification.	Vision Transformer (ViT) (best model), DenseNet121, VGG16, MobileNetV2.Patch-level accuracy: 99.9%, Generalization accuracy: 78.8%	**(1)** Generalization accuracy lower than patch-level. **(2)** Model struggled with finer details across different tissue regions	Highlights ViT's potential for patch-based classification while emphasizing challenges in whole-slide analysis, reinforcing the importance of integrating both local and global tissue

Author(s) & Year	**Objective**	**Methodology**	**key Findings**	**Limitations**	**Relevance**
Saranyaraj et al.(2023) [5]	Early detection of breast cancer by classifying HER-2 and ER biomarkers from histopathology images.	**(1)** Applied Stacked Sparse Autoencoder (SSAE), for feature extraction. **(2)** Classified HER-2 and ER using deep Convolutional Neural Network (DCNN)	Accuracy: 99.73% for nuclei classification, HER-2 and ER Patch level classification accuracy: 97.52%.	**(1)** The study focuses only on HER-2 and ER biomarkers, ignoring other critical biomarkers like PR. **(2)** Lack of External Validation. **(3)** High computational cost.	Demonstrates effective classification but limited biomarker scope and computational inefficiency
Che et al.(2023) [6]	Develop an automated method for HER2 IHC scoring in whole-slide images (WSI) using deep learning	**(1)** Threshold based segmentation used to separate ROI. **(2)** Extracts tumor areas and segments WSIs into patches. **(3)** Uses ResNet34 for binary classification (tumor/non-tumor)	Patch-level precision: 95.77%, F1-score: 83.09%	Single biomarker (HER2) considered: Does not address ER and PR, limiting its clinical applicability.	Filter Area belong to Background. Demonstrates HER2 scoring techniques and patch-based segmentation but lacks comprehensive biomarker evaluation, which we incorporate in our approach.

Author(s) & Year	**Objective**	**Methodology**	**key Findings**	**Limitations**	**Relevance**
Shamai et al. (2024) [7]	Predict ER, PR, and HER2 receptor status from H&E-stained slides using deep learning. Improve diagnostic accuracy and reduce misdiagnoses.	**(1)** Trained separate models for ER, PR, and ERBB2. **(2)** Local Feature: CNNs **(3)** Global Feature Aggregates local features to represent overall tissue architecture.	AUC: 0.955 (ER), 0.807 (PR), 0.846 (ERBB2)	**(1)** No shared feature learning across receptor types. **(2)** Late fusion restricts adaptive learning of multi-scale relationships.	The need for a Parallel processing to preserves spatial relationships between local and global features.
Goyal et al. (2024) [4]	Predict breast cancer recurrence risk using WSIs and clinicopathologic data as an alternative to Oncotype DX test	**(1)** Sequential feature extraction pipeline **(2)**Local features are first extracted using CNNs. **(3)**Global features (tissue-wide structures) are then extracted using Vision Transformers	**(1)**Local and global features are processed separately, leading to potential loss of spatial relationship. **(2)**ER+, HER2 (focuses on recurrence risk prediction rather than direct biomarker classification	The model successfully differentiates broad tumor patterns but struggles with capturing subtle interactions between local cell features and overall tissue structure.	Highlights the need for parallel feature extraction to prevent loss of spatial relationships, better capture biomarker expression patterns.

Author(s) & Year	**Objective**	**Methodology**	**key Findings**	**Limitations**	**Relevance**
Mujawar et al. (2025) [8]	To improve breast cancer classification in whole-slide images (WSIs) using stacked convolutional neural networks (CNNs)	**(1)**Used patch-based classification (224×224 pixels) followed by stacked CNNs with larger patch sizes (512×512, 768×768, 1024×1024) for improved accuracy. **(2)**Feature extraction from dense probability maps was used for WSI-level classification with Random Fores	Binary classification: 90% accuracy Three-class classification: 76.6% accuracy.	**(1)**Lower performance in three-class classification due to limited contextual information in small patches. **(2)**High computational cost for larger input sizes like 1024×1024.	This paper provides insights into whole-slide image classification using CNN-based models, highlighting the impact of patch size and stacked architectures on performance.
Aboudessouki et al. (2023) [9]	Develop a deep learning pipeline for automated classification of HER2 biomarker in breast cancer histology images	**(1)**The pipeline extracts patches (128×128) from WSIs, applies preprocessing (Otsu thresholding), and classifies using DenseNet for HER2 status. **(2)**Provides both patch-wise and pixelwise classification.	Patchwise classification:0.97 (Accuracy), Pixelwise classification:0.79 (Accuracy)	**(1)**Lower accuracy in pixelwise classification **(2)**Evaluated on a small dataset (10 cases). **(3)** Only classifies HER2 biomarker (not a full molecular subtyping model yet)	Uses Otsu thresholding for background removal, which helps focus on relevant tissue regions.

Author(s) & Year	**Objective**	**Methodology**	**key Findings**	**Limitations**	**Relevance**

Table 1: Summary of Related Work in Breast Cancer Detection

2.2. Parallel Feature Learning VS Standalone Models

Deep learning models for breast cancer detection can be designed to operate as standalone architectures, sequential or parallel feature learning techniques. Standalone models, such as CNN-based or ViT-based approaches, focus on extracting either local or global features independently. In contrast, parallel feature learning methods integrate multiple architectures to enhance both local and global feature representation. The following table outlines the differences between these approaches, highlighting their key characteristics in medical image analysis.

Author(s) & Year	Objective	Model used	Methodology	Results	Conclusion
khan et al (2024) [10]	To develop a deep learning model for classifying breast cancer categories (benign and malignant subtypes) from histopathological slides using a global-local feature extraction approach.	**G**LNET (Global–Local CNN-based informed model)	**Parallel Approach:** **(1)**Global Feature Extractor (GFE): Extracts whole-slide features (color, shape, tissue size) using ResNet101. **(2)** Local Feature Extractor (LFE): Extracts region-of-interest features using ResNet101. **(3)** Integrated global-local features into a single feature vector for classification.	**(1)Proposed Parallel GLNET** BreakHis Dataset Accuracy: 92.32% (100×) ICIAR2018-BachChallenge Dataset Accuracy: 90.33% (1000X) **(2) Comparison with Other Models** (BreakHis Dataset): ResNet101: 83.00% (100×) DenseNet201: 70.00% (100×) VGG19: 72.00% (100×) VGG16: 75.00% (100×)	The GLNET model uses a parallel approach to extract global and local features, enhancing feature learning for benign and malignant tumor classification and outperforms standalone models, achieving higher accuracy across datasets and magnification levels.

Author(s) & Year	**Objective**	**Model used**	**Methodology**	**Results**	**Conclusion**
Fiaz et al (2024) [11]	Propose a deep fusion-based vision transformer (DFViT) model that combines CNN and Vision Transformers to improve breast cancer classification	**D**FViT (Deep Fusion-based Vision Transformer)	**Parallel Approach:** **(1)**Uses CNN (VGG16) and Vision Transformer (ViT) simultaneously. **(2)**Fuses features extracted from CNN (local) and ViT (global) before classification. **(3)** Uses a Multi-Layer Perceptron (MLP) for final prediction	**(1)Proposed Parallel CNN-ViT (DFViT):** BACH dataset: 94% UC dataset: 84.56% BreakHis dataset: 95.29% **(2) Single CNN Models:** VGG16 BACH dataset: 77% ResNet (UC dataset): 70.1% DenseNet UCdataset: 61.5% **(3) Single ViT Model:** DeconViT BACH dataset: 79% UC dataset: 81.36%	Experiments show parallel fusion achieves higher accuracy than standalone CNN or ViT models.

Author(s) & Year	Objective	Model used	Methodology	Results	Conclusion
Wang et al (2023) [12]	Breast cancer histopathological image classification to detect the presence of metastatic tissue using a hybrid CNN-ViT model that enhances both local and global feature extraction.	**L**ocal-Global Vision Transformer (LGViT)	**Parallel Approach:** **(1)**CNN stem first extracts low-level local features. **(2)**ViT processes global features in parallel using self-attention. **(3)** Local-Global Multi-head Self-attention (LGMSA) bridges the local CNN features with global ViT representations.	LGViT achieved the highest accuracy (91.81%). **Comparison with key models:** ConvNeXt-T (90.15%) ResNet34 (88.98%) ViT-S (84.73%) Swin-T (84.99%)	The hybrid CNN-ViT approach improved both local and global feature extraction, leading to superior accuracy

Author(s) & Year	Objective	Model used	Methodology	Results	Conclusion
Kanadath et al (2024) [13]	To propose a CNN-ViT-based hybrid model with skip connections to improve histopathology image classification.	**C**ViTS-Net (Hybrid CNN-ViT model)	**Parallel Approach:** **(1)** Combines CNN and ViT to extract local and global features. **(2)**Introduced Multi-Scale Feature (MSF) Block using atrous convolutions. **(3)**Used Dual Global-Local Feature (DGL) Block combining depthwise convolutions and multi-head attention.	**CViTS-Net Achieved:** Chaoyang dataset: 96.06%, Lymphoma dataset: 99.61%, IDC dataset: 95% **Comparison with Other Models:** ResNet-34 (CNN): 78.07% XceptionNet (CNN): 84.39% ViT (Transformer): 58.5%–85.65% (various versions)	Demonstrates superior generalization on multiple datasets.

Author(s) & Year	Objective	Model used	Methodology	Results	Conclusion
Zhao et al (2024) [14]	To classify benign and malignant breast nodules using a hybrid CNN-Transformer model.	**A**ResNet-ViT (Hybrid CNN-Transformer)	**Parallel Approach:** **(1)**Dual-branch architecture: CNN for local features & ViT for global features, processed independently. **(2)**Attention mechanisms in CNN (Segmentation Mask Attention &Channel Attention). **(3)**Feature fusion via concatenation and MLP, followed by Sigmoid activation for classification.	**AResNet-ViT Achieved:** ACC: 88.9% **Comparison with Other Models:** ResNet18(CNN): 79.6% ViT(Transformer): 83.5% AResNet (CNN with attention): 85.7% ResNet-ViT: 86.3%	Effective combination of ARestNet and ViT Outperforms existing models.

Author(s) & Year	**Objective**	**Model used**	**Methodology**	**Results**	**Conclusion**
Hayat et al (2024) [15]	Classify breast cancer histopathology images (binary and multi-class) using a hybrid model.	**E**fficient-NetV2 (Small, Medium, Large)& Vision Transformer (ViT)	**Parallel Approach:** **(1)**Efficient-NetV2 extracts local fine-grained features. **(2)**ViT captures global contextual relationships.	**Efficient-NetV2-Large & ViT (Best Model)** Accuracy: 99.83% (binary), 98.10% (multi-class) **Comparison with Other Models:** EfficientNetV2-L Accuracy: 96.89%(Binary), 94.58% (Multiclass) Vision Transformer (ViT) Accuracy: 95.32%(Binary), 92.87% (Multiclass)	Hybrid model significantly improves accuracy over standalone CNN or ViT

Author(s) & Year	**Objective**	**Model used**	**Methodology**	**Results**	**Conclusion**
Abimoulo-ud et al (2024) [16]	To explore the combination of CNN and Vision Transformers (ViT) for breast cancer classification in histopathological images.	**V**iT-CNN	**Parallel Approach:** **(1)**Uses ViT-CNN hybrid models for feature extraction and classification. **(2)** Splits input images into patches for ViT processing. **(3)** Uses CNN layers for feature extraction before feeding into the attention mechanism of ViT. - Evaluates binary (benign vs. malignant) and multiclass (8 subtypes) classification.	**ViT-CNN** Binary Classification Accuracy: 98.64% Multiclass Classification Accuracy: 94.80% **Comparison with Other Models:** Compact Convolutional Transformer (CCT): Binary Classification Accuracy: 96.99% Multiclass Classification Accuracy: 84.60% MobileViT (MVIT) Binary Classification Accuracy: 97.52% Multiclass Classification Accuracy: 87.84%	ViT-cnn performed best for both tasks, with MVIT slightly behind and CCT lowest.

Table 2: Comparison of Standalone and Parallel Feature Learning Approaches in Medical Image Analysis

In medical image analysis, parallel feature learning consistently outperforms standalone architectures by integrating both local and global features, leading to improved accuracy in classification tasks. CNN-based models effectively capture fine-grained spatial details but struggle with long-range dependencies, while Vision Transformers (ViTs) excel at modeling global context but often lose critical local structures. Parallel feature learning addresses these limitations by combining both feature types, resulting in more robust and context-aware representations. This approach has demonstrated success across various fields of medical image analysis, reinforcing the need for its adoption in breast cancer biomarker classification. A comprehensive understanding of histopathological features is

essential for accurate ER, PR, and HER2 assessment, and parallel architectures provide a more holistic and clinically relevant analysis compared to standalone models.

2.3. Parallel vs. Sequential Architectures in Medical Image Analysis

Parallel and sequential architectures represent two distinct methodologies in medical image analysis, each designed to handle feature extraction and representation differently. Sequential architectures process features in a stepwise manner, where local and global information is extracted in separate stages, allowing for structured learning but potentially limiting seamless integration. In contrast, parallel architectures extract multiple feature types simultaneously, enabling concurrent learning of spatial and contextual information. These approaches have been explored across various medical imaging tasks, including classification, and disease detection, with differing implications for feature fusion and model interpretability. Several recent studies have examined the performance of these architectures in different medical imaging domains, highlighting their strengths and limitations.

Zeng et al.[17] proposed FUST, a parallel CNN-Transformer fusion framework for microscopic hyperspectral image classification. Their model integrates CNNs for multiscale spatial feature extraction and Transformers for spectral dependencies in a parallel setup. Compared to the Sequential CNN-Transformer (SST) approach, FUST demonstrated superior classification performance across multiple datasets. Notably, on the MN Dataset, FUST achieved an overall accuracy (OA) of 96.35%, compared to SST's 95.42%. Similarly, on the WBCs Dataset, FUST attained an OA of 98.60%, outperforming SST's 97.28%. These improvements, ranging from 1–3% in classification accuracy, highlights the advantages of parallel fusion in exploiting complementary spatial-spectral features.

In breast cancer histopathological image classification, Chitta et al.[18] introduced a hybrid model combining EfficientNetB7 and ViT-S16 in a parallel setup to classify benign and malignant tumors. EfficientNetB7 extracted local spatial features, while ViT-S16 captured global contextual information. The parallel approach achieved an accuracy of 96.83% on the BreakHis dataset, outperforming standalone CNN (95.36%), ViT-S16 (90.7%), and a sequential CNN-ViT model (94.38%). The findings emphasize the effectiveness of parallel CNN-Transformer architectures in enhancing classification performance in histopathological analysis.

A systematic review by Kim et al. [19] analyzed hybrid Vision Transformer (ViT)-CNN architectures for radiological image analysis, comparing parallel and sequential designs. The review examined 34 studies from 2020–2024 and found that parallel architectures, such as TECNN and CPT U-Net, outperformed sequential models in classification and restoration tasks, with an AUC of 0.834 and an F1-score of 0.968. The study concluded that parallel architectures are particularly well-suited for classification tasks, as they enable joint decision-making by integrating CNN and Transformer-based feature learning.

Finally, Hu et al. [20] introduced MambaConvT, a hybrid medical image classification model combining CNN, State Space Models (SSM), and Transformers. The model was evaluated on four diverse medical imaging datasets, including ISIC 2018 (dermatoscopic images), Kvasir (endoscopic images), CPN-CX (chest X-rays), and brain tumor MRI scans. MambaConvT, which incorporates parallel CNN, SSM branches and Transformer-based fusion, outperformed sequential architectures such as CNN-T and SSM-T. The proposed

model achieved 97.38% accuracy on the CPN-CX dataset, further reinforcing the benefits of parallel architectures in preserving fine-grained details and long-range dependencies.

Collectively, these studies demonstrate that Parallel architectures provide significant advantages over sequential approaches in medical image analysis by enabling simultaneous extraction of spatial and contextual features, leading to improved classification accuracy, enhanced feature fusion, and better model interpretability. Unlike sequential methods, which process features in distinct stages, parallel architectures allow joint learning, preserving fine-grained details while capturing long-range dependencies, making them particularly effective in complex tasks such as classification, and disease detection. These benefits have been demonstrated across various medical imaging domains, including histopathology, radiology, dermatology, and endoscopy, where parallel models consistently outperform their sequential counterparts. In the context of breast cancer biomarker classification from whole-slide imaging (WSI), where precise feature extraction across different magnifications and spatial scales is critical, a parallel approach can enhance robustness and accuracy. By integrating CNNs for local feature extraction and Transformers for global contextual understanding, parallel architectures provide a more comprehensive analysis of biomarker expression, supporting their effectiveness in breast cancer diagnosis and reinforcing their suitability for this research.

Chapter 3

3.1 Gap Analysis

Existing deep learning approaches for breast cancer diagnosis face several limitations. CNN-based models struggle to capture long-range dependencies in tissue analysis, limiting their ability to understand broader structural patterns within whole-slide images (WSIs) [2]. On the other hand, Vision Transformer (ViT) models emphasize global feature extraction but may overlook fine-grained cellular structures that are critical for precise biomarker evaluation [3]. Additionally, many methods focus on individual biomarker classification, such as ER, PR, or HER2, without considering their combined effect, resulting in a less comprehensive diagnostic view [6]. Another key limitation is that most approaches train separate models for each receptor, preventing shared feature learning across biomarker types and reducing the potential for improved classification performance [7]. Furthermore, the sequential processing of local and global features can disrupt spatial relationships, leading to incomplete feature representation and suboptimal diagnostic accuracy [4]. These challenges highlight the need for an improved deep learning framework that integrates local and global feature extraction while preserving spatial coherence and enabling multi-biomarker classification within a unified pipeline.

3.2 Problem Statement

Current deep learning methods for breast cancer analysis struggle with the vast size of WSIs and sparsely annotated regions, leading to loss of tissue level information. CNNs struggle with capturing long-range dependencies [2], while ViTs emphasize global structures but miss fine-grained cellular details [3]. Many methods focus on individual biomarkers without shared feature learning [7], limiting diagnostic comprehensiveness. Furthermore, sequential processing of local and global features disrupts spatial relationships, reducing

model effectiveness [4]. So the problem at hand is to develop a parallel feature learning approach to integrate biomarker interactions dynamically and enhance diagnostic accuracy.

3.3 Research Questions

This study aims to address the following research questions:

3.3.1. Primary Research Question

1. How can a parallel feature learning approach improve the classification accuracy of ER, PR, and HER2 biomarkers in whole-slide imaging (WSI)?

3.3.2. Secondary Research Questions

1. What impact does shared feature learning have on biomarker interaction modeling compared to individual biomarker-based models?
2. How does parallel processing of local (CNN) and global (ViT) features influence spatial coherence and classification performance in breast cancer diagnosis?

3.4 Methodology

In recent years, advanced deep learning techniques have emerged as powerful tools for breast pathology analysis, particularly in biomarker classification. Rehman et al.[3] developed a deep learning-based two-stage classification framework for HER2-SISH slides, distinguishing between Normal, Amplified, and Non-Amplified regions. The first stage focused on patch-level classification, followed by whole-slide image prediction. While their approach achieved near-perfect accuracy (99.9%) at the patch level, whole-slide accuracy was significantly lower (78.8%), highlighting challenges in integrating local and global tissue information. Additionally, the study solely concentrated on HER2 and did not account for other critical biomarkers like ER and PR, limiting its applicability for comprehensive breast cancer characterization. Similarly Shamai et al. [7] attempted to address multi-biomarker classification by developing deep learning models for ER, PR, and HER2 status prediction from HE-stained slides. Their methodology employed CNNs for feature extraction and a global feature aggregation step to analyze tissue structure. Although the model performed well, with AUC scores of 0.955 for ER, 0.807 for PR, and 0.846 for HER2, it processed each biomarker separately, restricting the potential for shared feature learning. Moreover, their late fusion strategy for feature aggregation posed challenges in preserving spatial relationships between biomarkers, potentially reducing the model's ability to capture cross-biomarker dependencies.

To Overcome the existing limitations, this research introduces a deep learning framework for the classification of breast cancer biomarkers (ER, PR, and HER2) from whole-slide images (WSIs). The proposed model employs a shared feature extraction module that integrates both Convolutional Neural Networks (CNNs) and Vision Transformers (ViTs) in a parallel configuration. Unlike sequential processing approaches, this design allows the model to independently capture fine-grained local features (e.g., nuclear and cellular boundaries) using CNNs, while ViTs extract global contextual information related to tissue structure and spatial organization.To ensure effective representation learning, the extracted features from both branches undergo a feature fusion process, preserving spatial

relationships across different levels of histopathological detail. Finally, a classification head is employed to predict biomarker status (ER, PR, and HER2), enabling a comprehensive and robust analysis of breast cancer subtyping. This parallel feature extraction strategy aims to improve classification accuracy and robustness while preserving essential spatial relationships in WSI analysis.

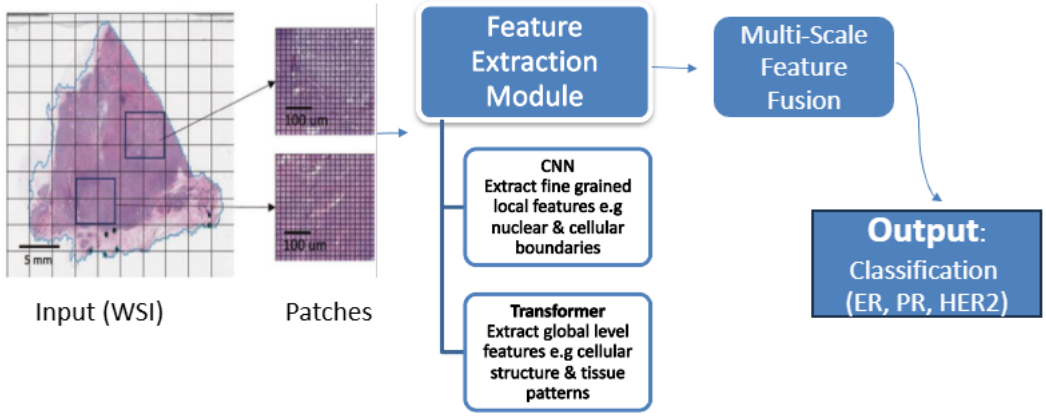


Figure 1: Methodology Overview

3.5 Dataset & Sources

This research acknowledges the availability of two publicly accessible datasets, the Post-NAT-BRCA and the Early Breast Cancer Core-Needle Biopsy (BCNB) datasets, alongside a proprietary dataset from the Pakistan Institute of Medical Sciences (PIMS).

Post-NAT-BRCA Dataset:

1. **Source:** Accessible through The Cancer Imaging Archive (TCIA) at https://www.cancerimagingarchive.net/collection/post-nat-brca/
2. **Structure and Attributes:** This dataset comprises 96 Hematoxylin and Eosin (HE) stained whole slide images (WSIs) from 54 patients with breast cancer. Each WSI is accompanied by clinical features and expert pathology annotations, including tumor cellularity assessments and cell type classifications. The images are stored in uncompressed .svs file format, scanned at 20x magnification using an Aperio slide scanner. Additionally an Excel (.xlsx) file contains clinical features for each patient, such as age, treatment details, and ER/PR/HER2 status. Each row in the spreadsheet corresponds to a single (.svs) slide.
3. **Ethical Considerations:** The dataset has been anonymized to protect patient identities, with all personal identifiers removed. Its use complies with ethical standards for research involving human subjects.
4. **Justification for Dataset Selection:** The Post-NAT-BRCA dataset provides high-resolution WSIs accompanied by detailed clinical and pathological annotations, including ER, PR, and HER2 status. This comprehensive information is essential

for developing and validating deep learning models aimed at assessing residual tumor cellularity and other pathological features in breast cancer patients.

BCNB Dataset:

1. **Source:** Available through the Grand Challenge platform at https://bcnb.grand-challenge.org/
2. **Structure and Attributes:** The BCNB dataset includes 1,058 HE stained WSIs from early breast cancer patients. Each image has been examined and annotated by two independent and experienced pathologists. The dataset also provides comprehensive clinical data for each patient, encompassing age, tumor size, tumor type, ER, PR, HER2 status, histological grading, molecular subtype, number of lymph node metastases, and the metastatic status of axillary lymph nodes (ALN). The WSIs are provided in .jpg format, with slides scanned using an Iscan Coreo pathologic scanner at 200x magnification.
3. **Ethical Considerations:** The BCNB dataset has been desensitized, with all personal identifiers removed, ensuring compliance with ethical standards for research involving human subjects.
4. **Justification for Dataset Selection:**The BCNB dataset is particularly relevant for this research as it provides a substantial number of WSIs accompanied by detailed clinical and pathological annotations, including ER, PR, and HER2 status. The availability of clinical data for these biomarkers allows for effective validation of the models' predictive capabilities.

PIMS Dataset:

While the Post-NAT-BRCA and BCNB datasets offer valuable resources, they have limitations in terms of detailed pathological annotations specific to certain biomarkers. To address these gaps, this research will incorporate breast pathology slides from the Pakistan Institute of Medical Sciences (PIMS) Pathology Department. Expert histopathologists will perform segmentation and provide comprehensive annotations, including detailed pathological features and classifications.

1. **Source:** Breast pathology slides from the PIMS Pathology Department.
2. **Justification for Dataset Selection:**Utilizing the PIMS dataset addresses the limitations of the publicly available datasets by providing detailed pathological annotations tailored to the specific requirements of this research. This approach ensures a comprehensive dataset that enhances the depth and applicability of the study.

3.6 Evaluation Strategy & Metrics

To ensure the validity and reliability of the proposed model, a comprehensive evaluation strategy will be employed, incorporating multiple performance metrics and comparative analyses against existing state-of-the-art methods.

Performance Metrics: The classification performance of the model will be assessed using the following key metrics:

1. **Accuracy:** Measures the overall correctness of predictions.
2. **Precision:** Evaluates the proportion of correctly predicted positive cases out of all predicted positives, crucial for reducing false positives.
3. **Recall (Sensitivity):** Assesses the model's ability to correctly identify all relevant cases, minimizing false negatives.
4. **F1-score:** Provides a balance between precision and recall, making it a robust measure for imbalanced datasets.

Justification for Metrics: Given the medical significance of biomarker classification, recall is particularly important to ensure minimal misclassification of positive cases. Additionally, the F1-score is selected as a primary metric to provide a holistic view of model performance, especially in scenarios with class imbalances.

Comparative Analysis: The proposed approach will be benchmarked against existing deep learning methods used for biomarker classification. This comparison will highlight improvements in classification accuracy and Performance. State-of-the-art CNN-based and transformer-based models will be considered as baselines for this evaluation.

3.7 Expected Outcomes

The anticipated outcomes of this research include:

1. **Improved Classification Accuracy:** By utilizing a parallel feature extraction mechanism, the model is expected to achieve higher accuracy than conventional CNN-only or transformer-only approaches.
2. **Enhanced Robustness:** The independent learning of local and global features is likely to improve model generalization, reducing errors in biomarker classification.
3. **Preservation of Spatial Relationships:** The integration of CNNs and ViTs ensures that both fine-grained cellular details and broader tissue structures are effectively utilized, leading to more reliable predictions.
4. **Comparative Superiority Over Baseline Models:** Experimental results are expected to demonstrate that the proposed framework outperforms existing state-of-the-art methods in terms of precision, recall, and F1-score.
5. **Potential Clinical Impact:** The research aims to provide a more reliable deep learning-based diagnostic tool that can assist pathologists in breast cancer subtyping, leading to more informed treatment decisions.

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